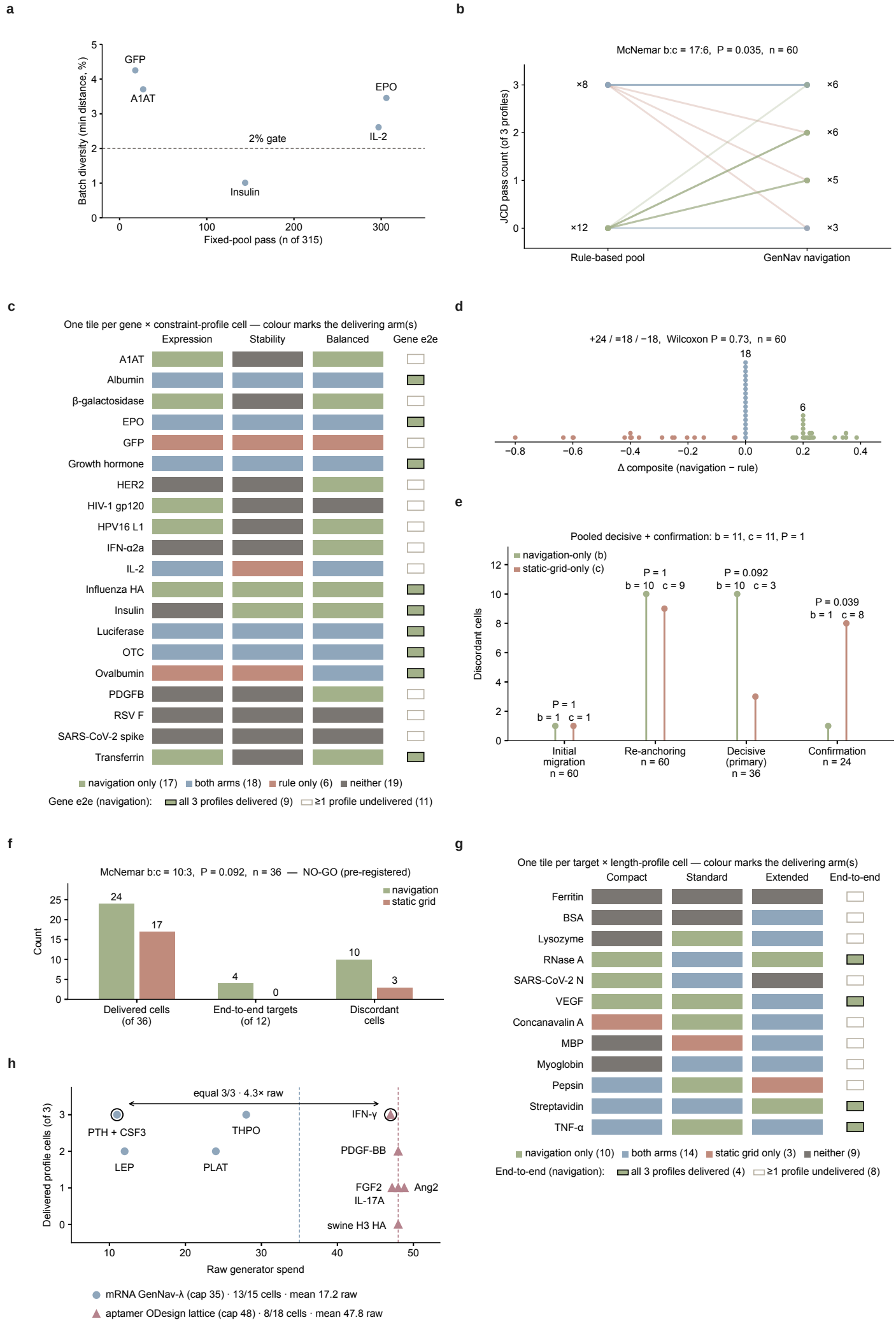

a
Batch diversity (min distance, %)
Fixed-pool pass (n of 315)
GFP
A1AT
EPO
IL-2
Insulin
2% gate
b
McNemar b:c = 17:6, P = 0.035, n = 60
JCD pass count (of 3 profiles)
Rule-based pool
GenNav navigation
×8
×12
×6
×6
×5
×3
c
One tile per gene × constraint-profile cell — colour marks the delivering arm(s)
Expression
Stability
Balanced
Gene e2e
A1AT
Albumin
β-galactosidase
EPO
GFP
Growth hormone
HER2
HIV-1 gp120
HPV16 L1
IFN-α2a
IL-2
Influenza HA
Insulin
Luciferase
OTC
Ovalbumin
PDGFB
RSV F
SARS-CoV-2 spike
Transferrin
navigation only (17)
both arms (18)
rule only (6)
neither (19)
Gene e2e (navigation):
all 3 profiles delivered (9)
≥1 profile undelivered (11)
d
+24 / =18 / −18, Wilcoxon P = 0.73, n = 60
Δ composite (navigation − rule)
e
Pooled decisive + confirmation: b = 11, c = 11, P = 1
navigation-only (b)
static-grid-only (c)
Discordant cells
P = 1
b = 1 c = 1
P = 1
b = 10 c = 9
P = 0.092
b = 10 c = 3
P = 0.039
b = 1 c = 8
Initial migration n = 60
Re-anchoring n = 60
Decisive (primary) n = 36
Confirmation n = 24
f
McNemar b:c = 10:3, P = 0.092, n = 36 — NO-GO (pre-registered)
navigation
static grid
Count
Delivered cells (of 36)
End-to-end targets (of 12)
Discordant cells
g
One tile per target × length-profile cell — colour marks the delivering arm(s)
Compact
Standard
Extended
End-to-end
Ferritin
BSA
Lysozyme
RNase A
SARS-CoV-2 N
VEGF
Concanavalin A
MBP
Myoglobin
Pepsin
Streptavidin
TNF-α
navigation only (10)
both arms (14)
static grid only (3)
neither (9)
End-to-end (navigation):
all 3 profiles delivered (4)
≥1 profile undelivered (8)
h
equal 3/3 · 4.3× raw
Delivered profile cells (of 3)
Raw generator spend
PTH + CSF3
THPO
IFN-γ
LEP
PLAT
PDGF-BB
FGF2
Ang2
IL-17A
swine H3 HA
mRNA GenNav-λ (cap 35) · 13/15 cells · mean 17.2 raw
aptamer ODesign lattice (cap 48) · 8/18 cells · mean 47.8 raw